```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# nano math_ops.sh
```

```
root@kali: /home/aman_singhal
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  GNU nano 8.7.1                    math_ops.sh *
#!/bin/bash

echo "Enter first number:"
read a

echo "Enter second number:"
read b

x=$a
y=$b

# GCD calculation using Euclidean algorithm
while [ $y -ne 0 ]
do
    temp=$y
    y=$((x % y))
    x=$temp
done

gcd=$x
lcm=$(( (a * b) / gcd ))

echo "GCD of $a and $b is: $gcd"
echo "LCM of $a and $b is: $lcm"
```

```
GNU nano 8.7.1                    math_ops.sh *

echo "Enter a number to check for prime:"
read n

flag=0

if [ $n -le 1 ]; then
    flag=1
fi

for ((i=2; i<=n/2; i++))
do
    if [ $((n % i)) -eq 0 ]; then
        flag=1
        break
    fi
done

if [ $flag -eq 0 ]; then
    echo "$n is a Prime Number"
else
    echo "$n is NOT a Prime Number"
fi
```

```
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┌──(root㉿kali)-[/home/aman_singhal]
└─# chmod +x math_ops.sh
```

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# ./math_ops.sh
Enter first number:
12
Enter second number:
18
GCD of 12 and 18 is: 6
LCM of 12 and 18 is: 36
Enter a number to check for prime:
13
13 is a Prime Number

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```